

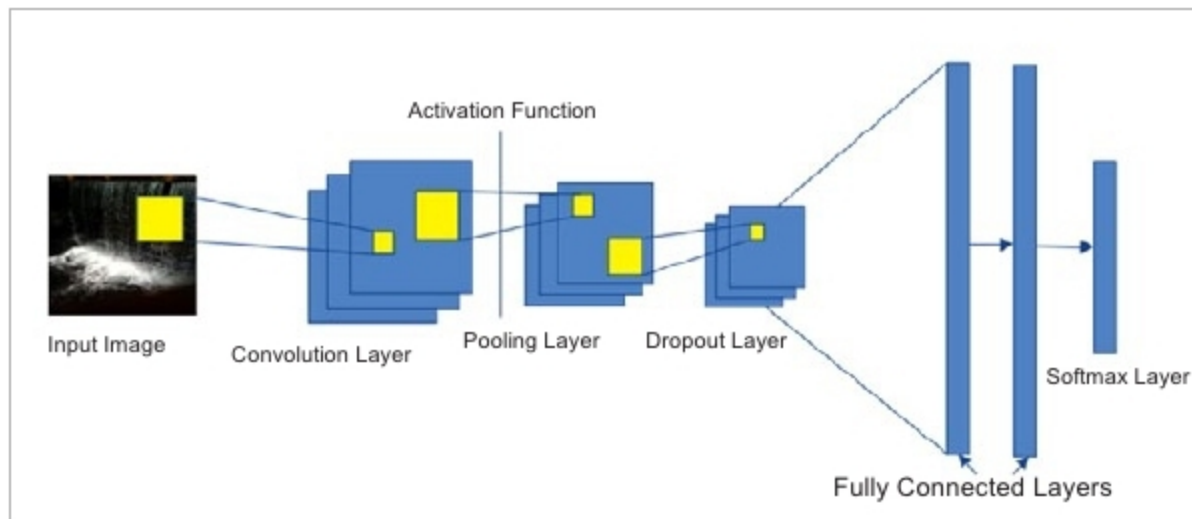


Figure 2: Illustration of CNN model architecture

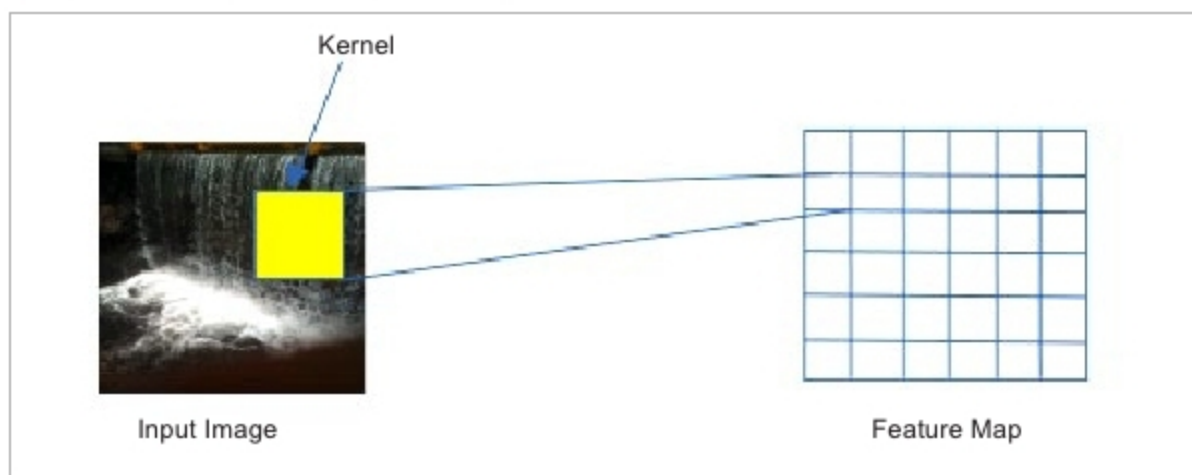


Figure 3: Convolution operation

Convolution layer

The convolution layer is the most important building block of the CNN architecture. The neurons in the first convolution layer are not connected to all the pixels in the input image, but are connected to pixels of their local receptive fields, as shown in Figure 2. Mathematically, the convolution layer merges two sets of information to produce a feature map. It applies the convolution filter or kernel to the input and produces the feature map. Let us understand this better with an example.

Let us suppose that the input image size is 6×6 and the kernel size is 3×3 . To draw a parallel to the visual cortex, the visual field size is 6×6 and the local receptive field size is 3×3 . To generate the feature map, the kernel is made to slide over each pixel location of the input image. Hence a single value is generated for each convolution operation (the summation or multiplication of the pixel value with the corresponding kernel value

in the same location). The area on the input image that is used during each convolution operation is the receptive field, the size of which is equivalent to the kernel size. Since the kernel is made to slide over the entire input image, an equal-sized feature map as the input image is produced after the convolution operation is applied to the entire image. The same procedure is followed for 3D images or multi-colour channel input images. An independent feature map is also produced for each channel in that case. Since there is one feature map for each convolution operation using a kernel, for a normal 2D image in grey scale, the number of feature maps produced in each layer is equal to the number of kernels used at that layer. For 3D images, it is three times the number of kernels due to three colour channels.

During a convolution operation, for each row x and column y in the feature map, the value at position (x, y) will be the total of the multiplication of values

in rows x to:

$x + kh - 1$, columns y to $y + kw - 1$

...where kh and kw are the height and width of the receptive fields. Hence, to produce a feature map equal to the input image size, suitable padding has to be done around the input image. In general, zero padding is applied to the input image.

It is also possible to slide the kernel over the input image by spacing out the pixel location. That means, as described above, the kernel is made to slide over each pixel location of the input image to generate the feature map. The distance between each successive receptive field is one (based on pixel locations). The spacing out or distance between two successive receptive fields is called the stride. As the kernel is made to slide over rows and columns, the stride can be different for sliding across rows (vertical direction) and columns (horizontal direction). With the stride value more than one, it is possible to produce a feature map that is smaller than the input size. With the stride, the neuron value in the feature map located at (x, y) is produced by rows.

$(x) * sh + kh - 1$, columns $(y) * sw + kw - 1$,

...where sh and sw are horizontal and vertical strides.

Each location (x, y) in the feature map will be a neuron generated using the specific kernel in that convolution layer. The neuron's weight at a location (x, y) will hence be a value, after the convolution operation is applied to rows x to:

$x + kh - 1$, columns y to $y + kw - 1$,

...where kh and kw are the height and width of the receptive field of the input.

The CNN model is built to learn the equation $WX + b$, where W is the weight of the neurons, X is the